



MTH634: Quiz 3



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Question # 8 of 10

SOLVED BY: **WAQAR HUSSAIN**

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{3\}, \{4\}, \{3, 4\}, X\}$ be a topology on X , then which of the following is true?

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



The set $\{\emptyset, \{3\}, \{4\}, X\}$ is an open cover of the set $\{4\}$



The set $\{\emptyset, \{3\}, \{4\}\}$ is an open cover of the set $\{4\}$



All of them

(Module No.117)



The set $\{\emptyset, \{4\}\}$ is an open cover of the set $\{4\}$



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Question # 9 of 10

SOLVED BY: **WAQAR HUSSAIN**

Every topological space is first countable.

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



False

A **topological space** is first countable if there is a countable neighborhood base (or local base) at each of its points.



True





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Question # 9 of 10

SOLVED BY: **WAQAR HUSSAIN**

Every topological space is first countable.

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



False



True

(Module 112)



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Question # 10 of 10

SOLVED BY: **WAQAR HUSSAIN**

Which of the following statement is false?

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



Discrete topology on a countable set X is second countable



Any finite set with any topology is second countable



Discrete topology on a countable set X is second countable



The set $\mathbb{R}$ with usual topology is second countable

According to Second Countable:

1- Discrete topology on a countable set X is second countable

2- Any finite set with any topology is second countable

3- The set $\mathbb{R}$ with usual topology is second countable





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Question # 10 of 10

SOLVED BY: **WAQAR HUSSAIN**

Which of the following statement is false?

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



Discrete topology on a countable set X is second countable



Any finite set with any topology is second countable



Discrete topology on a real line $\mathbb{R}$ is second countable

(Module 114)



The set $\mathbb{R}$ with usual topology is second countable



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Question # 1 of 10

SOLVED BY: WAQAR HUSSAIN

Metric topology induced by $d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$ is called

Select the correct option

PRESENTED BY: TALEEMI MARKAZ



usual topology



discrete



indiscrete



None of them

Example 1:

Recall trivial metric on a set X , i.e.

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

Where $x, y \in X$. We know that the topology generated by the set of all single tons is **discrete topology**. So, the metric topology induced by the above metric is the **discrete topology** on X .





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Question # 1 of 10

SOLVED BY: **WAQAR HUSSAIN**

Metric topology induced by $d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$ is called

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



usual topology



discrete topology

(Module No.107)



indiscrete topology



None of them



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Question # 2 of 10

SOLVED BY: **WAQAR HUSSAIN**

Metric topology induced by $d(x, y) = |x - y|$ on $\mathbb{R}$ is called

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



usual topology



discrete topology



indiscrete topology



None of them

Example 1:

Recall $\mathbb{R}$ with usual metric

$$d(x, y) = |x - y|$$

Where $x, y \in \mathbb{R}$. We know that the topology generated by the set of all open intervals is **usual topology**. So, the metric topology induced by the above metric is the **usual topology** on $\mathbb{R}$.





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Question # 2 of 10

SOLVED BY: **WAQAR HUSSAIN**

Metric topology induced by $d(x, y) = |x - y|$ on $\mathbb{R}$ is called

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



usual topology

(Module No.107)



discrete topology



indiscrete topology



None of them



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Question # 3 of 10

SOLVED BY: **WAQAR HUSSAIN**

Which of the following statement is true?

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



All spa



All spa



Metrizable Space:

A topological space (X, τ) is called metrizable space if there exists a metric d on X that induces the topology τ .

Remark: Not all spaces are metrizable.





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Question # 3 of 10

SOLVED BY: **WAQAR HUSSAIN**

Which of the following statement is true?

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



All spaces are metrizable



All spaces are not metrizable

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Question # 4 of 10

SOLVED BY: **WAQAR HUSSAIN**

Every metric space is first countable.

Select the correct option

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True

Recall that:

1- Every metric space is first countable.

2- $\mathbb{R}$ with usual topology is also first countable



False





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Question # 4 of 10

SOLVED BY: **WAQAR HUSSAIN**

Every metric space is first countable.

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



True

(Module No.112)



False



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Question # 5 of 10

SOLVED BY: **WAQAR HUSSAIN**

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$ be a topology on X , then the local base (B_x) of the point $x = 1$ is _____

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



$\{\{1\}, \{1, 2\}, X\}$



$\{\{1\}, \{2\}\}$



$\{\{1\}, \{2\}\}$



None of

SOLUTION:

$$X = \{1, 2, 3, 4, 5, 6\}$$

$$\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$$

Local Base of point $x = 1$

Check subspace of τ where 1 is present.

$$1 \in \{1\}, \quad 1 \in \{1, 2\}, \quad 1 \in X$$

So

Local base of point $x = 1$ is $\{\{1\}, \{1, 2\}, X\}$





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Question # 5 of 10

SOLVED BY: **WAQAR HUSSAIN**

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$ be a topology on X , then the local base (B_x) of the point $x = 1$ is _____

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



$\{\{1\}, \{1, 2\}, X\}$

(Module 114, Conceptual)



$\{\{1\}, \{2\}, \{1, 2\}, X\}$



$\{\{1\}, \{2\}, X\}$



None of them



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Question # 6 of 10

SOLVED BY: **WAQAR HUSSAIN**

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$ be a topology on X , then the local base (B_x) of the point $x = 2$ is _____

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



$\{\{1\}, \{1, 2\}, X\}$



$\{\{2\}, \{1, 2\}, X\}$



$\{\{1\}, \{2\}, X\}$



None of them





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Question # 6 of 10

SOLVED BY: WAQAR HUSSAIN

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$ be a topology on X , then the local base (B_x) of the point $x = 2$ is _____

Select the correct option

PRESENTED BY: TALEEMI MARKAZ



$\{\{1\}, \{1, 2\}, X\}$



$\{\{2\}, \{1, 2\}, X\}$



$\{\{1\}, \{2\}, X\}$



None

SOLUTION:

$$X = \{1, 2, 3, 4, 5, 6\}$$

$$\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$$

Local Base of point $x = 2$

Check subspace of τ where 2 is present.

$$2 \in \{2\}, \quad 2 \in \{1, 2\}, \quad 2 \in X$$

So

Local base of point $x = 2$ is $\{\{2\}, \{1, 2\}, X\}$





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Question # 6 of 10

SOLVED BY: **WAQAR HUSSAIN**

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$ be a topology on X , then the local base (B_x) of the point $x = 2$ is _____

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



$\{\{1\}, \{1, 2\}, X\}$



$\{\{2\}, \{1, 2\}, X\}$

(Module 114, Conceptual)



$\{\{1\}, \{2\}, X\}$



None of them



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Question # 7 of 10

SOLVED BY: **WAQAR HUSSAIN**

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{3\}, \{4\}, \{3, 4\}, X\}$ be a topology on X , then which of the following is true?

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



The set $\{\emptyset, \{3\}, \{4\}\}$ is not an open cover of the set $\{2\}$



The set $\{\emptyset, \{3\}\}$ is not an open cover of the set $\{4\}$



None of the above



The set $\{\emptyset, \{3\}, \{4\}\}$ is an open cover of the set $\{4\}$

1- The set $\{\emptyset, \{3\}, \{4\}\}$ is not an open set of the set $\{2\}$
because the union of $\emptyset \cup \{3\} \cup \{4\} = \{3, 4\} \notin \{2\}$
2- The set $\{\emptyset, \{3\}\}$ is not an open cover of the set $\{4\}$
because the union of $\emptyset \cup \{3\} = \{3\} \notin \{4\}$
3- The set $\{\emptyset, \{3\}, \{4\}\}$ is an open cover of the set $\{4\}$
because the union of $\emptyset \cup \{3\} \cup \{4\} = \{3, 4\} \in \{4\}$
So, **option D** is right answer.





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Question # 7 of 10

SOLVED BY: **WAQAR HUSSAIN**

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{3\}, \{4\}, \{3, 4\}, X\}$ be a topology on X , then which of the following is true?

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



The set $\{\emptyset, \{3\}, \{4\}\}$ is an open cover of the set $\{2\}$



The set $\{\emptyset, \{3\}\}$ is an open cover of the set $\{4\}$



None of them



The set $\{\emptyset, \{3\}, \{4\}\}$ is an open cover of the set $\{4\}$

(Module No.117)



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Question # 8 of 10

SOLVED BY: **WAQAR HUSSAIN**

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{3\}, \{4\}, \{3, 4\}, X\}$ be a topology on X , then which of the following is true?

Select the correct option

PRESENTED BY: **TALEEMI MARKAZ**



The set $\{\emptyset, \{3\}, \{4\}, X\}$ is an open cover of the set $\{4\}$



T

1- The set $\{\emptyset, \{3\}, \{4\}, X\}$ is an open set of the set $\{4\}$ because the union of $\emptyset \cup \{3\} \cup \{4\} \cup X = \{1, 2, 3, 4, 5, 6\} \in \{4\}$



A

2- The set $\{\emptyset, \{3\}, \{4\}\}$ is an open cover of the set $\{4\}$ because the union of $\emptyset \cup \{3\} \cup \{4\} = \{3, 4\} \in \{4\}$



The set $\{\emptyset, \{4\}\}$ is an open cover of the set $\{4\}$

3- The set $\{\emptyset, \{4\}\}$ is an open cover of the set $\{4\}$ because the union of $\emptyset \cup \{4\} = \{4\} \in \{4\}$

All are correct, So, **option C** is right answer.

